Assignment – 2.10

DHCP Configuration

Q - 1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

- **DHCP Server Configuration:**
- **Network Setup:**

  DHCP Server: 192.168.1.1

  PC1: Gets IP automatically → 192.168.1.2

  PC2: Gets IP automatically → 192.168.1.3

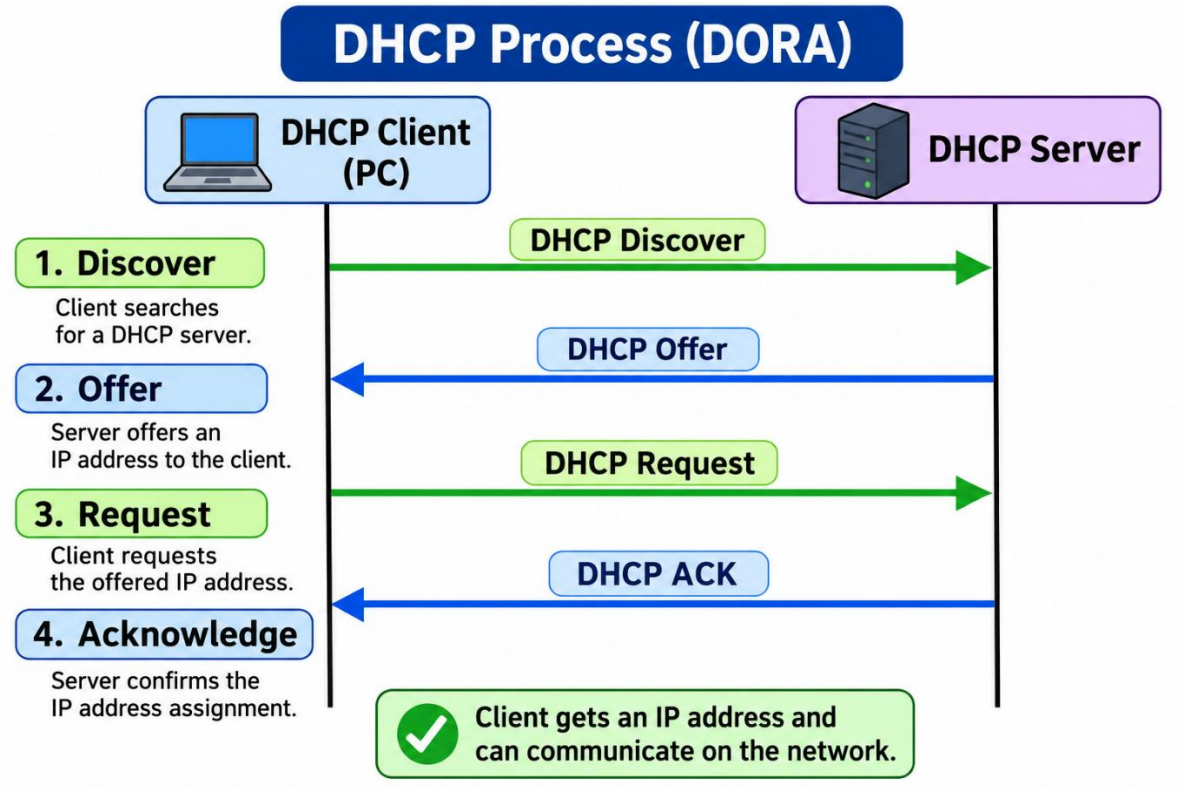
- **Verification:**

  Both PCs were set to DHCP, and they received different IP addresses automatically from the DHCP pool.

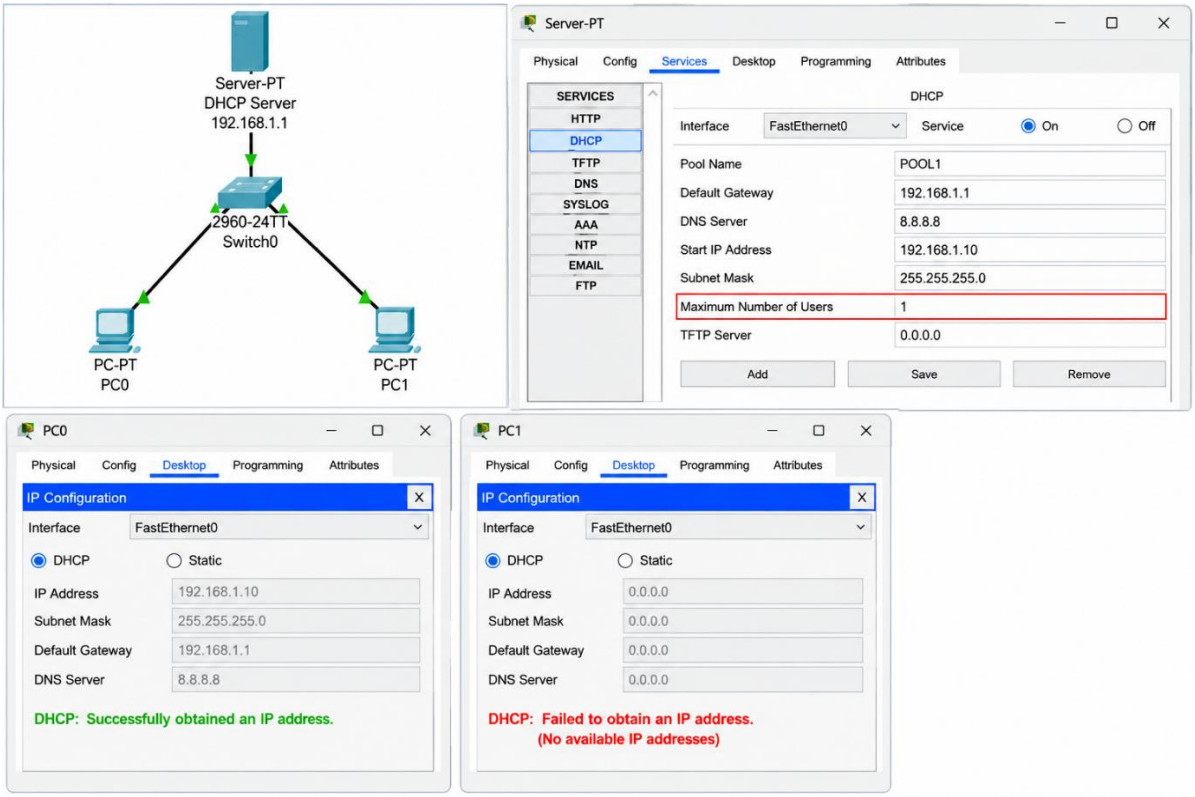
- **Result:**

  DHCP was configured successfully and assigned unique IP addresses to both client PCs.

Q - 2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage. Hint: Use arrows to show the direction of each message between client and server.



Q - 3. Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.



➢ **Explanation:**

Only one IP address is available in the DHCP pool, so only one PC can receive an IP address. The second PC fails to get an IP because the DHCP server has no free IP addresses left.

Q - 4. Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

- **DHCP in Restaurant Wi-Fi:**
- **Assigned IP Addresses:**

Device	IP Address
Laptop	192.168.1.100
Smartphone	192.168.1.101

- **Why is DHCP useful?**

  DHCP automatically gives IP addresses to devices connected to restaurant WiFi. Customers do not need to enter IP addresses manually. It saves time and avoids IP conflicts.

- **Result:**

  Both laptop and smartphone successfully get different IP addresses from the DHCP server.